

BIOL3203 Food Microbiology

Damian Lau

2025-2026 Semester 1

Introduction

This course begins with a discussion of marine biology, covering the Introduction to Food Microbiology, Foodborne Pathogens, Detection and Enumeration of microbes in foods, Spore and their significant, Beneficial Bacteria, Factors that influence microbes in food and Physical Method of Food Preservation. This course aims to introduce the microscopic of food. Upon completion, students will be familiar with the mechanism and phenomenon in food microbiology.

Contents

Introduction	1
1 Lecture 7 Factor Affecting Microbial Growth I	3
2 Lecture 8 Factor Affecting Microbial Growth II	20
3 Lecture 9 Physical Methods of Food Preservation	29

1 Lecture 7 Factor Affecting Microbial Growth I

1. Microbial Growth in Food

- Uncontrolled Growth in food
- May cause **Spoilage**
 - (a) Loss of Food Quality
 - (b) Example : Slime, Gas, Off-flavors
 - (c) Unspoiled : Foodborne Illness can still exist and invisible
- May cause Foodborne Illness : Food pathogens reaching infectious dose and produce toxins

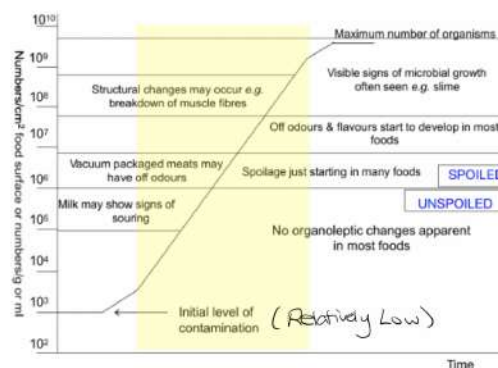


Figure 1: Microbial Growth Level

2. Lag Phase

- Cells adjust to the new environment
 - (a) **Inducing / Suppressing Enzyme Synthesis and Activity**
 - (b) **Initiating DNA Replication**
 - (c) Differentiating into vegetative cells
 - ⇒ Spore Formers = Can form endospores
- Length of Lag Phase depends on **Temperature & Inoculum Size & Physiological History of the Organism**
 - ⇒ Larger Inoculum → Shorter Lag Phase
 - ⇒ Refrigeration (Low Temperature) = Longer Lag Phase

- Goal of Food Preservation
 - (a) Extend the Lag Phase as long as possible
 - (b) Once enter Log Phase → Food reach **Spoilage Threshold** → Unsafe / Spoiled

3. Log Phase

- Binary Fission Growth
 - (a) Double Bacteria Population at **Equal time Intervals**
 - (b) Equal Time Intervals = Doubling Time (t_d) / Generation Time (t_{gen})
 - (c) Doubling Time : Inversely related to **Specific Growth Rate μ**
 $\Rightarrow \downarrow$ Doubling Time, $\uparrow$ Microorganisms Growth

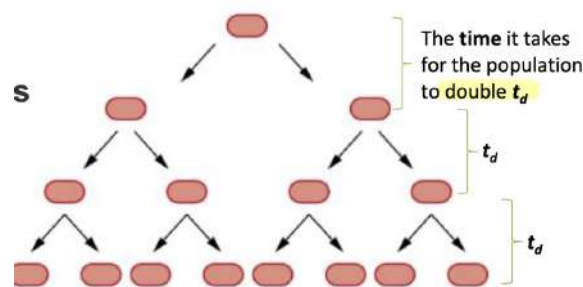


Figure 2: Binary Fission

- For any time (t), Number of Organisms (N) is **Directly Proportional** to the Initial Number of Organisms (N_0)
- μ = **Specific Growth Rate**
- Relationship of **Time & Number of Cells** :

$$N = N_0 e^{\mu t}$$

$\Rightarrow N, N_0$ Unit : CFU/g

$\Rightarrow t$ Unit : Hours (h)

$\Rightarrow \mu$ Unit : Per Hours (h^{-1}) $\Rightarrow$ Used to Predicting Microbial Population Size & Quantifying Microbial Growth Rates

- Relationship is **Exponential**
- Time of a microbial population to reach a certain level :

$$\Delta t = \frac{2.3 \log_{10}(\frac{N}{N_0})}{\mu}$$

$$\Rightarrow 2.3 = \frac{1}{\log_{10} e}$$

$\Rightarrow \Delta t$ = Time Interval between **two measurement points**

- Doubling Time vs Generation Time :

$$t_d = \frac{0.693}{\mu}$$

$$\Rightarrow \text{Assumption : } \frac{N}{N_0} = 2$$

$$\Rightarrow 0.693 = 2.3 \log_{10}(\frac{N}{N_0})$$

$$\Rightarrow t_d \text{ Unit : Hours (hr)}$$

4. Temperature is not needed in the equation

$\Rightarrow$ But still have great influence over the final cell number

$\Rightarrow$ Low Temperature = Low Growth Rate

5. Factor affecting Microbial Growth

(a) Intrinsic Factors : Characteristic Inherent to the Food **Itself**

- Water Activity
- pH
- Oxidation-reduction Potential
- Nutrient Content
- Antimicrobial Constituents
- Biological Structure

(b) Extrinsic Factors : Characteristic External to the Food

- Temperature
- Relative Humidity
- Gaseous Environment

6. Water Activity (A_W)

- Microbial Growth : Controlled by **Availability of Water**
 $\Rightarrow$ Not the amount of water in the food
- Moisture Content $\neq$ Water Activity
- Water in Food
 - (a) Free Water
 - Unbound Water
 - Available for **Chemical Reaction & Microbial Growth**
 - (b) Bound Water
 - Bound by **Ionic Interaction, Hydrogen Bonding, Capillary Action in Food Microstructure**
 - Ionic Interaction Example : Can bind to salt
 - Hydrogen Bonding Example : Can bind to sugar
 - Capillary Action Example : Water trap in food's physical structure
 - Microbes cannot use bound water
 - Can bound to carbohydrates
 - Mainly occurs inside foods
- **Used to measure the energy status of water in foods**
 - Scale : 0 = No Free Water - 1 = Pure Water
 - Higher A_W = Higher Energy = More Available Water
 - Formula :

$$a_W = \frac{P}{P_0}$$
 - P = Vapor Pressure of Water in Food
 - P_0 = Vapor Pressure of **Pure Water at the Same temperature**
 - Equilibrium Relative Humidity (ERH) = Gas Phase Water Molecules Efficiency of the air surrounding the food in a sealed container
 - $\Rightarrow$ Bound Water cannot escape into gas phase
 - $\Rightarrow$ Escaped Water $\rightarrow$ Vapor Pressure

– Formula Interchanging :

$$a_w = ERH\%$$

- Optimal Water Activity for Microbes : **0.97-0.99**
 $\Rightarrow \downarrow a_w \rightarrow \downarrow \text{Growth Rate}$
- **No Growth below a_w minimum = 0.60 (Fixed Temperature : 25°C)**
 $\Rightarrow$ Not Necessarily died
 $\Rightarrow$ Reversible if the water activity is above minimum level again
- **Gram Negative Bacteria need higher a_w than Gram Positive Bacteria**
 $\Rightarrow$ Mold need the least
- **Unique Pathogenic Bacteria : *Staphylococcus aureus* Minimum $a_w = 0.86$**
- Halophilic = Salt-Loving
- Xerotolerant = Dry-Tolerating
- Xerophilic = Dry-Loving
- Osmophilic = Sugar-Loving

Group of microorganisms	Minimal a_w required
Most bacteria	0.91-0.88
Most yeasts	0.88
Regular molds	0.80
Halophilic bacteria	0.75
Xerotolerant molds	0.71
Xerophilic molds and osmophilic yeasts	0.62-0.60

Figure 3: Water Activity in Different Microorganisms

7. Water Activity of Foods

- Water Activity : A Predictor of **Microbial Stability**
- Food $a_w \geq$ Microbes Minimum a_w = Have Growth in food

Typical a_w of various foods	
Foods	a_w
Fresh, raw fruits, vegetables, meat, and fish	>0.98
Cooked meat, bread	0.95-0.98
Cured meat products, cheeses	0.91-0.95
Sausages, syrups	0.87-0.91
Rice, beans, peas	0.80-0.87
Jams, marmalades	0.75-0.80
Candies	0.65-0.75
Dried fruits	0.60-0.65
Dehydrated vegetables, spices, milk powder	0.20-0.60

Support growth of most bacteria, yeasts, and molds

No microbial growth

*Reprinted from Farkas J, p 547-591, in Doyle MP, Beuchat LR, Montville TJ, ed, Food Microbiology: Fundamentals and Frontiers, 2nd ed, ASM Press, Washington, DC, 2001

Figure 4: Water Activity & Foods

8. Reduce Water Activity Methods

- Adding Salt / Sugar
 - (a) Binding with free water
 - (b) Binding Water Efficiency : Salt > Sugar
- Dehydration
 - (a) Removing Available Water
 - (b) Example : Hot Air Drying, Freeze-drying
 - (c) Can provide lowest water activity of other method
- Freezing : Converting Free Water → Ice Crystal

9. pH

- Measurement of **Acidity / Alkalinity** of a food
- pH Formula

$$pH = -\log[H^+]$$

- Formula can measure the concentration of Hydrogen Ions

10. Microbial growth and Metabolism are affected by pH

- (a) Acidic Environment : Affects the **activity & stability** of macro-molecules
 - ⇒ Example : Enzyme Functions, Cellular Activities
- (b) Internal pH of Microbial Cells : Neutral
 - ⇒ Most microbes does not affect by External pH
 - ⇒ Maintain Stable Internal pH : Use Energy
 - ⇒ Use energy to maintain pH → Less energy for growth & toxin production

11. pH vs Microbial Growth

(a) Optimal pH range = Maximum Growth Rate : 1-2 pH units

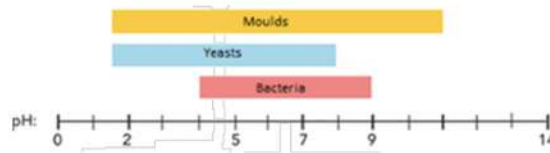


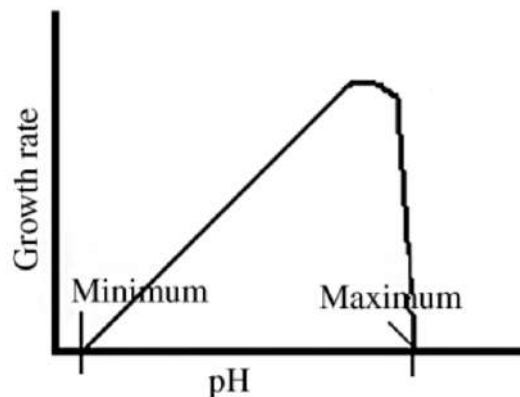
Figure 5: Optimal pH Range

(b) Most microbes grow best pH : Neutral

(c) **Bacteria** Optimal pH : pH 6.0 – 8.0
 $\Rightarrow$ Acid-producing bacteria : pH 5.0 – 6.0

(d) **Yeast** Optimal pH : 4.5 – 6.0

(e) **Mold** Optimal pH : 3.5 – 5.0



Plotting microbial growth rate against pH produces a bell-shaped curve

Figure 6: Microbial Growth Rate against pH

12. Methods of Maintaining Intracellular pH against Environmental pH Changes

(a) Homeostatic Response : Use this when External pH > 6.0
 $\Rightarrow$ Cell use **Proton Pump** $\rightarrow$ Expel H^+ from Cytoplasm
 $\Rightarrow$ Always functioning

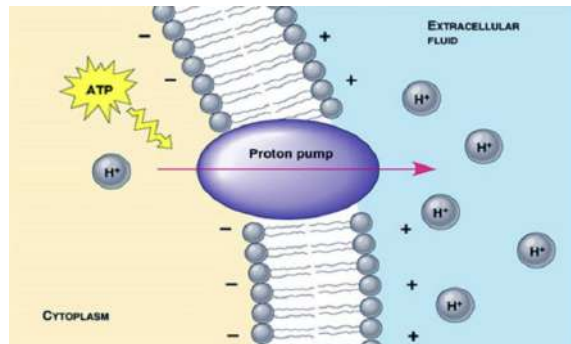


Figure 7: H^+ Exchange

- (b) Acid Tolerance Response (ATR) : Use this when **External pH 5.5 – 6.0**
 - ⇒ Cell synthesizes **ATR-induced Protein** + Use **Membrane-bound ATPase Pump** → Maintain **pH > 5**
 - ⇒ Inducible : Trigger by moderate external pH
- (c) Acid Shock Protein : Use this when **External pH 3.0 – 5.0**
 - ⇒ Cell synthesizes **Regulatory Proteins** → Survive severe acid stress
 - ⇒ Triggered by several low external pH

13. pH of Foods

- (a) Most food pH value : < 7
- (b) **Food Safety and Regulation Critical Limit pH : ≤ 4.6**
 - ⇒ *Clostridium botulinum* cannot grow / produce neurotoxin
- (c) Acid Foods : pH ≤ 4.6
 - ⇒ Pathogen do not grow / Grow very slow
 - ⇒ E.g. : Apples, Fermented Food, Fruit Juice, Tomato
- (d) Low-acid Foods : pH > 4.6
 - ⇒ Good growth for bacterial pathogen and spoilage microbes (Molds, Yeast)
 - ⇒ Low-acid foods require **Intensive Thermal Processing** → Ensure Food Safety
 - ⇒ E.g. : Most Vegetables, Cheese, Meat, Fish, Egg

(e) Food with Low pH : More Microbiologically Stable than Neutral Food



Figure 8: Different pH Foods

14. Methods of Controlling pH

(a) ↑ Acidity of Food by **Fermentation**

- Production of Lactic Acid → ↓ pH
⇒ Acid-producing bacteria : Lactic Acid Bacteria

(b) Strong Acids

- Cannot Enter Cell Membrane
- Only lower External pH
- E.g. : Hydrochloric Acid (HCl), Phosphoric Acid
- HCl : Dissociate H^+ & Cl^- **Completely**

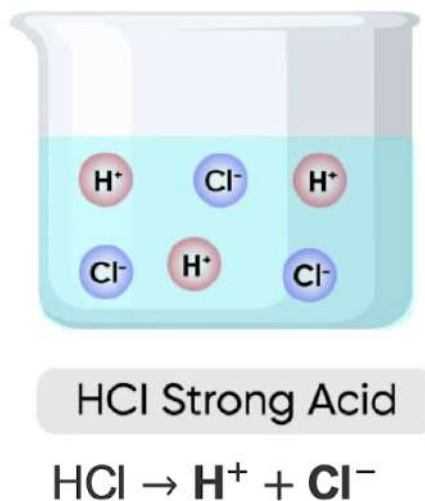
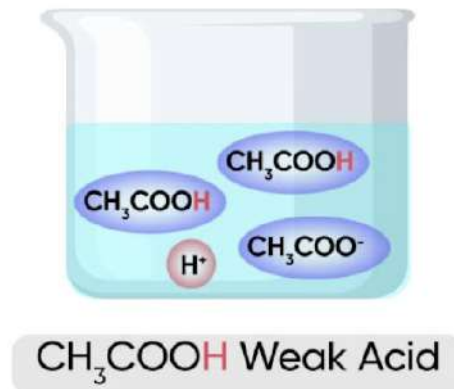


Figure 9: HCl Strong Acid Dissociation

(c) Weak Acids

- More Effective Inhibitor : Can enter cells more easily

Figure 10: CH₃COOH Dissociation

- E.g. Organic Acids, Acetic Acid, Lactic Acid, Citric Acid
- CH₃COOH dissociated H⁺ & CH₃COO⁻ very little

(d) Weak acids : An Effective Preservative

- Partial Dissociation of Weak Acids : Establish equilibrium with **Dissociated & Undissociated Forms**
- E.g. $\text{CH}_3\text{COOH} \rightleftharpoons \text{H}^+ + \text{CH}_3\text{COO}^-$
 - Dissociated form : CH₃COO⁻
 - Charged + Cannot Pass Through Cell Membrane
 $\therefore$ Cell Membrane is Hydrophobic Lipid Bilayer $\rightarrow$ Charged Material cannot pass
 - Undissociated Form : CH₃COOH
 $\Rightarrow$ Uncharged $\rightarrow$ Free to permeates the cell membranes
 $\rightarrow$ Enter the Cell

(e) Weak Acids Enter Cytoplasm (In Undissociated Form)

- Dissociate inside the cytoplasm : **Neutral pH > Outside pH**
 $\Rightarrow$ That is **Releasing H⁺ in Cytoplasm** $\rightarrow$ **$\downarrow$ Cytoplasmic pH**
- Maintain Internal pH : Expelling H⁺
 $\Rightarrow$ That is **ATP Consumption** $\rightarrow$ **Growth Slows Down**

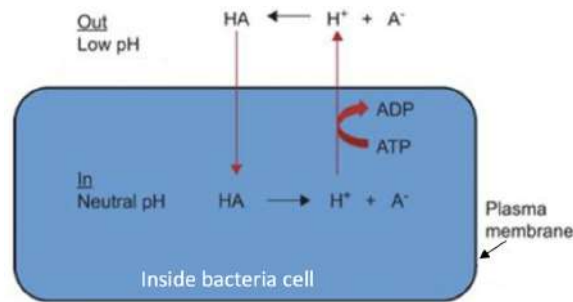


Figure 11: H^+ Exchange

- Bacteria wants to maintain neutral pH

15. Oxidation-Reduction Potential (Redox)

- Living cells generate energy through redox reactions
 $\Rightarrow$ Electrons transfer to Electron Acceptor
- E.g. Aerobic Respiration $\rightarrow$ Oxygen = Final Electron Acceptor
- Redox Reaction Occurs when there is **transfer of electron between Atom / Molecules**

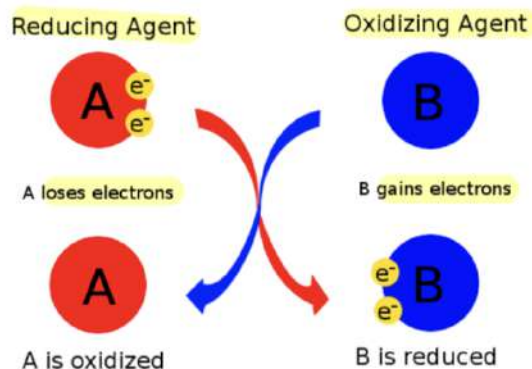


Figure 12: Redox Reaction

16. Redox Potential : Denoted as E_h

- $\Rightarrow$ The tendency of a substrate to **Accept / Donate Electrons**
- $\Rightarrow$ Food Redox Potential = Growth Capability of microbes

17. Unit : Millivolts (mV)

18. Positive E_h : +mV

- Oxidizing Environment
- Oxygen is present = Aerobic

19. Negative E_h : -mV

- Reducing Environment
- No Oxygen = Anaerobic

20. Oxygen : Strong Oxidizing Agent (High Redox Potential)
∴ Microbe Oxygen Requirement = Redox Potential

21. Microbes vs Redox Potential

(a) Aerobes

- Use Free Oxygen
- Oxygen Detoxifying Enzymes : Catalase + Superoxide Dismutase (SOD)
- Redox Potential (E_h) for Growth : +500 – +300mV
- Most Molds and Yeast
- Examples : *Bacillus*, *Pseudomonas*

(b) Facultative Anaerobes

- Prefer using Oxygen but can grow without Oxygen
- Oxygen Detoxifying Enzymes : Catalase + Superoxide Dismutase (SOD)
- Redox Potential (E_h) for Growth : +300 – -100mV
- Examples : *Salmonella*, *E. coli*, *Staphylococcus aureus*

(c) Anaerobes

- Oxygen is toxic to these microbes
- No Oxygen Detoxifying Enzymes
⇒ Radicals accumulate and damage the cells
- Redox Potential (E_h) for Growth : +100 – < -250mV
- Examples : *Clostridium spp.*

(d) Toxic Effects of Oxygen on Anaerobes

- Oxygen Metabolism : Produce **Reactive O₂ Radicals**
⇒ E.g. Superoxide Radicals, Hydrogen Peroxide (H₂O₂)
- Aerobes & Facultative Anaerobes can tolerate **Reactive O₂ Radicals** by **Oxygen Detoxifying Enzymes**
- Oxygen Detoxifying Enzymes : Superoxide Dismutase (SOD)
⇒ Converting Superoxide Radicals (Strong Toxin) to Hydrogen Peroxide (Weaker Toxin)
- Catalase : Decomposing Hydrogen Peroxide (Weaker Toxin) to Oxygen (Safe)

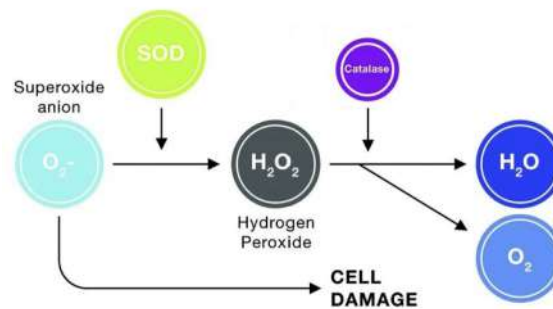


Figure 13: Toxification on anaerobes

22. Redox Potential of Foods

(a) Chemical Composition

- Mainly look at the reducing substances
⇒ We can get a lower Redox Potential
- Example : Ascorbic Acid, Reducing Sugar, -SH group of Proteins (A protein that contain Sulfur)

(b) pH of Food

- Mainly look at how pH decrease (Same as H⁺ Increase)
⇒ We can get a higher Redox Potential
- Example : Fruits vs Raw Meat

(c) Processing

- Grinding = Break down the food physically

- Increase the exposure of food to air
 $\Rightarrow$ We can get a higher Redox Potential
- (d) Storage Atmosphere (Exposure to air, Air = O₂)
 - Vacuum Packing, Canning $\rightarrow$ Remove O₂
 - Exclusion of air
 $\Rightarrow$ We can a lower Redox Potential
- (e) Microbial Growth
 - Microbes Consume Oxygen + Release Reducing Compound
 - We can get a lower Redox Potential

Food	E_h (mV)	pH
Raw meat (post-rigor)	-200	5.7
Raw minced meat	+225	5.9
Cooked sausages and canned meats	-20 to -150	ca. 6.5
Wheat (whole grain)	-320 to -360	6.0
Barley (ground grain)	+225	7.0
Potato tuber	ca. -150	ca. 6.0
Spinach	+74	6.2
Pear	+436	4.2
Grape	+409	3.9
Lemon	+383	2.2

Figure 14: Redox Potential of Foods

23. Plant Food (Fruits)

- High E_h : +300 – +400mV
- Common Reason of Food Spoilage : Aerobic Bacteria and Mold

24. Intact Raw Meat

- High E_h : -200mV
- Anaerobes bacteria : Decomposition of deep tissues (Center of the meat)
 $\Rightarrow$ Surface Spoilage : Contact to air $\rightarrow$ +ve E_h Potential

25. Nutrient Content : Use Nutrient to Synthesize Cellular Components and Energy

(a) Water

(b) Source of Energy

- Carbohydrates, Alcohols, Amino Acid, Lipids
- Most microbes prefer **Simple Sugars (Glucose)**
- Some accept **complex carbohydrates**
⇒ Can degrade it
⇒ E.g. Starch / Cellulose → Simple Sugar

(c) Source of Nitrogen

- Amino Acids, Nucleotides, Peptides (Proteins)
- Most microbes use **Amino Acid**
- Some can utilize Peptides & Proteins
⇒ E.g. Peptides & Protein → Amino Acids

(d) Vitamins & Growth Factor : Small amount of Vitamins B

(e) Mineral : Can be traced → Coenzymes

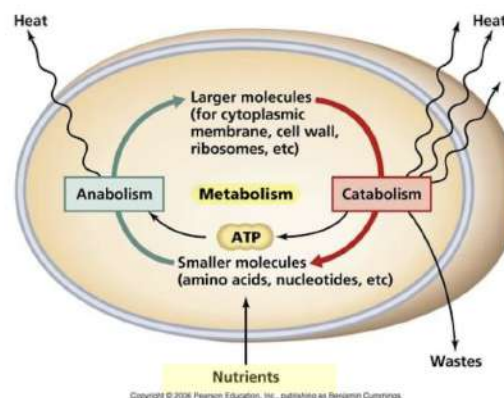


Figure 15: Metabolism

26. Nutrient Content vs Microbial Growth

- **Nutrient Amount Requirement : Molds < Gram-Negative Bacteria < Yeasts < Gram-Positive Bacteria**
⇒ Mold = Least Nutritionally Demanding
⇒ Gram-Positive Bacteria = Need More Nutrition

- E.g. Gram-positive Bacteria : Cannot make Vitamins B
⇒ Need to take it from food
⇒ Fruits cannot support as Vitamins B is low
- Microbial Growth Rate is limited by Availability of Nutrients
⇒ Most foods contain enough nutrients for most microorganisms
⇒ Hard to predict microbial growth by Only base on Nutrient Composition

27. Antimicrobial Constituents

- Natural Antimicrobial Compounds Food : Plant Sources, Egg, Cow Milk
- Plant Sources
 - Fruits, Vegetables, Tea
⇒ Contain Phenolic Compounds
⇒ E.g. Caffeic Acids, Ferulic Acid, Chlorogenic Acids
 - Spices : Essential Oil
⇒ E.g. Garlic (Allicin), Thyme (Thymol), Cloves & Cinnamon (Eugenol)
- Egg
 - Lysozyme = Can Hydrolyzes Bacterial Cell Walls
- Cow Milk
 - Lysozyme
 - Lactoferrin : Bacteria Cannot Use
 - Lactoperoxidase System : Can Kill Bacteria

28. Biological Structure

- Natural Covering Food : Provide Physical Barrier
- Function : Provide Protection against Microbes Damage
- Example : Seed Coat, Outer Covering of Fruits, Shell of Nuts, Shell of Eggs

- Physical Damage : Allow microbes to enter the food
⇒ Fruits skin damage → Undergo Faster Spoilage than not damaged

29. Interaction of all Intrinsic Factor = Can Check Overall Stability of Food
⇒ Can Predict the General Microorganisms willing to grow on certain food or not

2 Lecture 8 Factor Affecting Microbial Growth II

1. Extrinsic Factor : Characteristic External to the Food

- Temperature
- Relative Humidity
- Gaseous Environment

2. Temperature

(a) Microorganisms can grow in a wide range of temperatures

(b) We want to know microbe's **Minimum, Optimal, Maximum Growth Temperature**

⇒ Related to **Metabolisms & Enzymatic Reaction**

(c) $T < T_{\text{Optimal}}$

- Growth Rate : Slow
- Enzymatic Reaction : Slow
- Membrane Fluidity : Low → ↓ Nutrient Transport through membrane

(d) $T < T_{\text{Minimum}}$: Growth is stopped

(e) $T > T_{\text{Optimal}}$

- Growth Rate : Decline Sharply
- Protein & Enzymes : Denaturation Irreversible
- Cytoplasmic Membrane : Break

(f) $T > T_{\text{Maximum}}$: Vegetative Cells are died

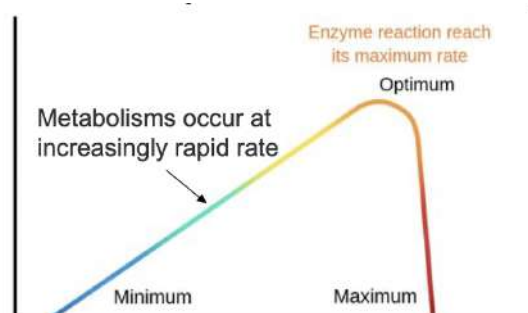


Figure 16: Metabolism Curve

3. Temperature vs Microbial Growth

- **Mesophiles** : Many foodborne pathogens
- **Psychrotrophs** : Can grow at refrigeration temperature
 ⇒ In a slow rate, But still grow
 ⇒ Spoilage of chilled food

Bacteria	Temperature(°C)		
	Min	Optimum	Max
Psychrophiles	-5 to 5	12 to 15	15 to 20
Psychrotrophs	-5 to 5	25 to 30	30 to 35
Mesophiles	5 to 15	30 to 45	35 to 47
Thermophiles	40 to 50	55 to 75	60 to 90

Figure 17: Bacteria vs Temperature

4. Low Temperature

- Maintain Membrane Fluidity by **Homeoviscous Adaptation**
 ⇒ ↑ **Unsaturated Fatty Acid** → Prevent Solidification of Cell Membrane
- Accumulate Compatible Solutes
 ⇒ E.g. Amino Acids, Beaine
 ⇒ Can reduce water freezing point → Preventing Intracellular Ice Crystal Formation
- Produce **Cold Shock Protein (CSPs)**
 ⇒ Function : Stabilize mRNA at Low Temperature

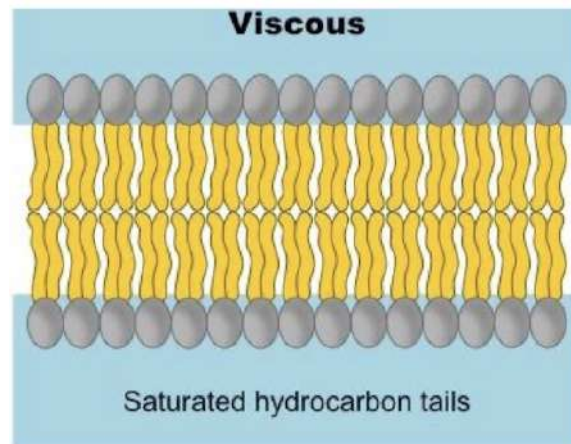


Figure 18: Viscous Cell membrane

- (d) Microbes will slow and stop growth during temperature decreases
- (e) Refrigeration : **Inhibits Mesophiles**
⇒ Cannot prevent **Psychrotrophs**, But grow slowly
- (f) Freezing : No Growth

5. High Temperature

- (a) Maintain Membrane Fluidity by **Homeoviscous Adaptation**
⇒ ↑ **Saturated Fatty Acid** → Reduce Cell Membrane Fluidity
- (b) Produce **Heat Shock Protein (HSPs)**
⇒ Function : Prevent Protein Denaturation

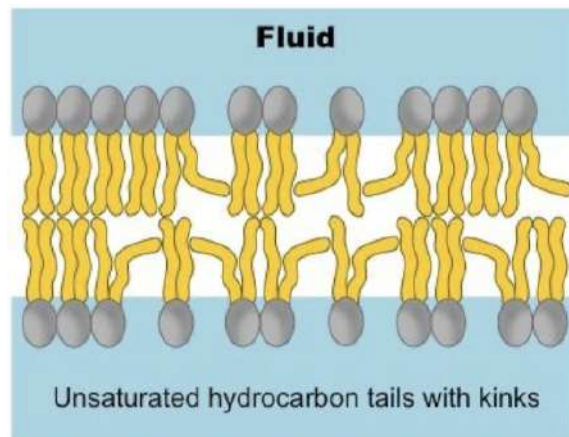


Figure 19: Fluid Cell Membrane

(c) Destruction of Microorganism : $\uparrow$ Temperature, $\uparrow$ Death Rate

(d) Pasteurization : Mild Heat = Kill Vegetative Cells

(e) Sterilization : High Heat = Kill Spores

6. Dangerous Temperature : $0^{\circ}\text{C} - 55^{\circ}\text{C}$



Figure 20: Microbial Growth vs Temperature

7. Relative Humidity (RH%)

(a) Measurement of **Amount of Water Vapor in Air**

(b) Compare to **Maximum amount of water vapor in the air could hold** at the same temperature

- (c) RH of storage environment can change the water activity (a_w) of food
 $\Rightarrow a_w = ERH\%$ = Equilibrium Relative Humidity
- (d) Any food can undergo **surface spoilage** $\rightarrow$ Stored under low RH Condition
 $\Rightarrow$ Prevent Migration of water passing into the food

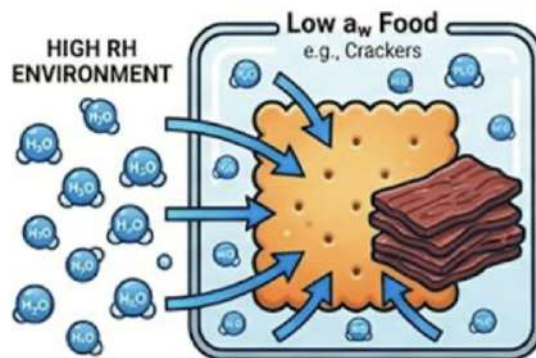


Figure 21: Migration of Water Molecules

- (e) Low a_w Food in High RH Environment
- **Have Surface Moisture Absorption**
 $\Rightarrow \uparrow a_w$
 - Microbial Growth
 $\Rightarrow$ Surface Mold Spoilage

8. Gaseous Atmosphere

- Atmosphere around the food = Presence of air affecting the **Redox Potential** (E_h)
 - Special Gases can Inhibit Microbes
- (a) Gases have **Direct Toxic Effect**
- CO₂ : Dissolved in Water $\rightarrow$ H₂CO₃ = Carbonic Acid = Weak Acid $\rightarrow$ Intracellular Acidification
 - O₃ : Generate Oxidizing Radicals
 $\Rightarrow$ **Highly Toxic to Anaerobes**
 $\therefore$ Oxygen is Toxic to Anaerobes $\Rightarrow$ **Can Inhibit Aerobes**

(b) Modifying Gas Composition : Use High CO₂ / Nitrogen
replace O₂ → ↓ O₂

⇒ **Help the Growth of Anaerobes**

⇒ E.g. *Clostridium Botulinum* ⇒ Refrigeration is still needed ⇒ **Can Inhibit Aerobes**

- Modified Atmosphere Packaging (MAP)
 - Change Food Packaging Gas Composition
 - Can use CO₂ or N₂ or O₂
 - ↑ Shelf Life of Food
 - ∴ Inhibit Food Spoilage

9. Thermal Destruction of Microbes

(a) Constant Temperature : Bacterial Thermal Destruction is **Exponentially Decrease** with time

(b) Formula

$$N = N_0 \times e^{-kt}$$

(c) N = Number of Viable Cell **After time (t) of Heating**

(d) N_0 = Initial Number of Viable Cells

(e) k = Rate Constant for destruction at temperature T
⇒ Unit : Time⁻¹

(f) t = Time

(g) Time Interval is fixed

(h) 10-Fold = 1 Log

- 1 Log Reduction = 90% Microbes are killed
- 2 Log Reduction = 99% Microbes are killed
- 3 Log Reduction = 99.9% Microbes are killed
- Concepts : Every Remaining has 90% microbes killed

(i) Time of Reducing Viable Bacteria by 10-Fold is Constant
∴ Related to Death Rate

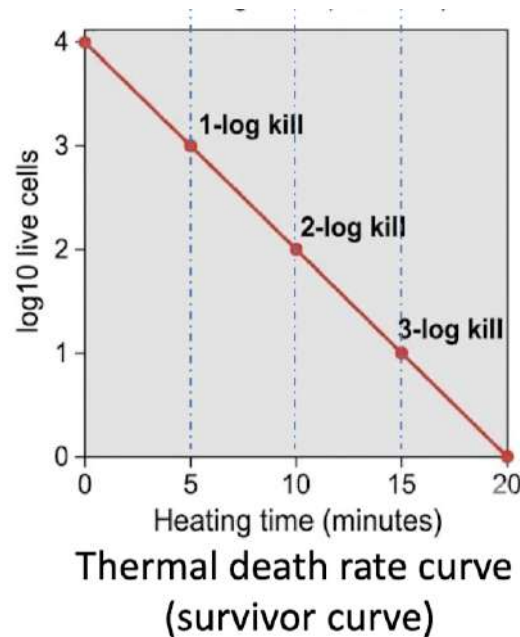


Figure 22: Survivor Curve

10. D Value = A measurement of time

- (a) Decimal Reduction Time (Fixed Specific Temperature)
- (b) Time = Period that used to kill microbes till 1 Log
- (c) Negative Reciprocal of Slope

$$D = \frac{-1}{\text{Slope}} = \frac{t_2 - t_1}{\log(N_0) - \log(N)}$$

- (d) Can used to check **Microbe's Heat Resistance at Specific Temperature**
- (e) $\uparrow D \rightarrow \uparrow$ Heat Resistance
 $\therefore D_{\text{Vegetative Cells}} < D_{\text{Spores}}$
- (f) **High Temperature = Small D Value**
 $\therefore$ Vegetative Cell die faster at high temperature

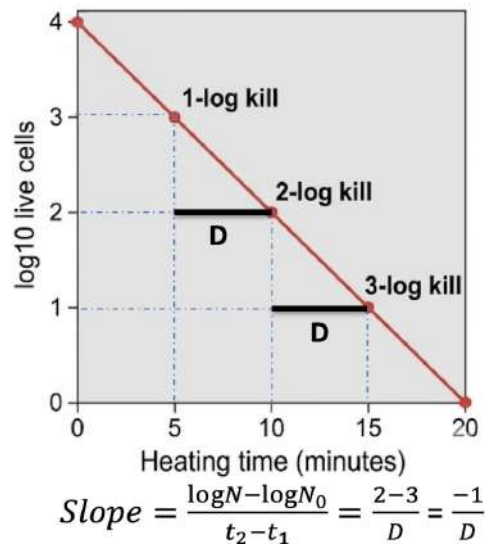


Figure 23: D Value Curve

(g) D Value vs Temperature : Logarithmic Relationship

11. Killing Rate : Increase when Temperature Increase

12. Z Value = A measurement of temperature

(a) Describes how **Lethality Changes with Temperature**

(b) Fixed Time

(c) Temperature used to reduce D Value : 1 Log

(d) Formula

$$Z = \frac{T_2 - T_1}{\log(D_1) - \log(D_2)}$$

(e) T_1 = Temperature at D_1

(f) T_2 = Temperature at D_2

(g) **↑ Z Value = Microbes are more resistant to Temperature Change**

(h) Formula in terms of D Value

$$D_2 = D_1 \times 10^{\frac{T_1 - T_2}{Z}}$$

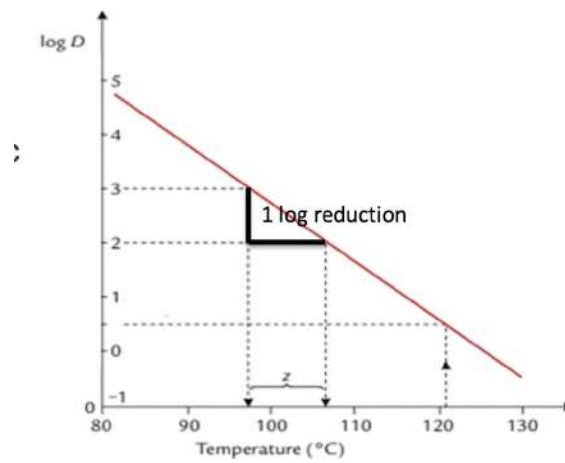


Figure 24: Thermal Death Time Curve

- (i) Thermal Death Time Curve = How D value changes at different temperature = Z Value

3 Lecture 9 Physical Methods of Food Preservation

1. Food Preservation

- Prevent **Deterioration & Spoilage**
- Extend Food Product shelf-life
- Without Changing Sensory Attributes
- Do not reduce nutritional composition

2. Aim of Food Preservation

- Kill Microbes
- Inhibit the Growth of Microbes
⇒ Add Minor Stresses
- Divided into Physical & Chemical Methods

3. Microorganisms Control

- Mode of Action
 - (a) Destruction : Cidal Effects
⇒ Killing of Microorganisms
 - (b) Inhibition : Static Effects
⇒ Slow Down / Stop the Growth of Microorganisms
 - (c) Separation : Removal of Microorganisms
⇒ Method : Filtration
⇒ Usually Used in Liquid / Gas Sample
- Action on Microbial Cells
 - (a) Damage Cell Wall
 - (b) Damage Plasma Membrane
 - (c) Damage Protein
 - (d) Damage Nucleic Acid

4. Factors Affecting Antimicrobial Effectiveness

- (a) Number of Microorganisms
 - ↑ Microbes Begin With → ↑ Eliminate time to the Population

(b) Exposure Time

- ↑ Food Exposure Time to Control Method → ↑ Number of Microbes would Killed
- Chemical Antimicrobials : Need Extended Exposure
⇒ Can Affect more Resistant Microbes / Endospores

(c) Microbial Characteristics

- Resistant Microbes → More Difficult to kill
- Endospores Formers vs Non Endospore Formers
- Gram Negative vs Gram Positive
- Enveloped Virus vs Non-enveloped Virus

5. Traditional Physical Methods for Preservation

- (a) Heat Treatments
- (b) Dehydration / Drying
- (c) Freezing
- (d) Changing The Atmosphere
- (e) Radiation

6. New Physical Method for Preservation

- (a) Dense-Phase Carbon Dioxide (DPCD)
- (b) Cold Plasma

7. Chemical Methods for Preservation

- (a) Naturally Occurring Food Antimicrobials
- (b) Artificial Food Antimicrobials

8. Heat Treatments

- Cheap & Effective Antimicrobial Method
- Medium Temperature > Optimum Growth Temperature :
Cell Multiplication Slow Down + Stop
- Additional Increase in Temperature : Membranes, Proteins, Ribosomes, Microbial Cells may have Injury / Death

- Mildly High Temperatures ($55^{\circ}\text{C} - 90^{\circ}\text{C}$) : Can Damage various cellular components
 ⇒ **Psychrophilic & Mesophilic Bacteria : Lose Activity**

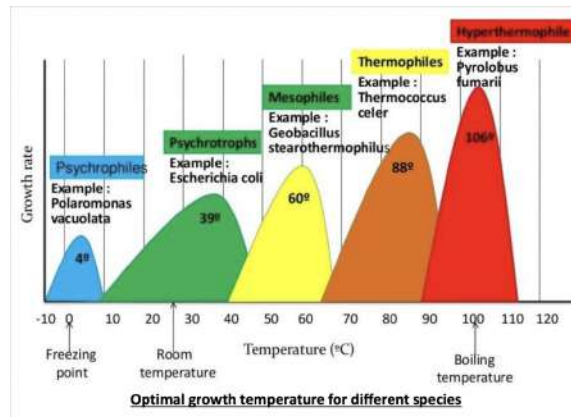


Figure 25: Optimal Temperature vs Microbes

9. **Temperature & Time** are the most significant factor for heat treatment
 ⇒ Protect food from pathogen
10. Effectiveness of Heat Sterilization Measurement
 - (a) High Temperature + Long Time : Not Necessary
 - (b) High Temperature + Short Time
 - (c) Low Temperature + Long Time
 - (d) Low Temperature + Short Time : Cannot Kill Microbes
11. Decimal Reduction Time (D-Value)
 - Measuring **Time (Minutes)** At a Give Fixed Temperature + **Decrease Viability by a factor of 10**
 ⇒ A Factor of 10 = 10-Fold = 1 Log
 - Used to find **Microbe Heat Resistance at a single temperature**
 - Can change with **The Strains & Characteristics of Suspending Medium**
 ⇒ E.g. Food Composition, Water Activity, pH
 ⇒ Low pH = Microorganisms are more sensitive to heat

- $\uparrow$ Processing Temperature $\rightarrow \downarrow$ D-value

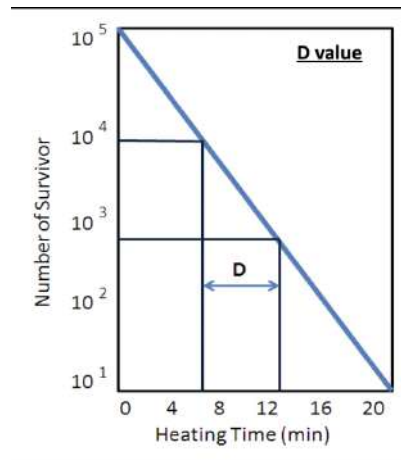


Figure 26: D-Value Curve

12. Thermal Resistance Constant (Z-Value)

- The degree required to **Change D-value by a factor of 10**
- Function 1 : Monitoring D-value
- Function 2 : How Lethality changes with temperature
- $\uparrow$ Z-Value, $\uparrow$ Resistance to High Heating Temperature

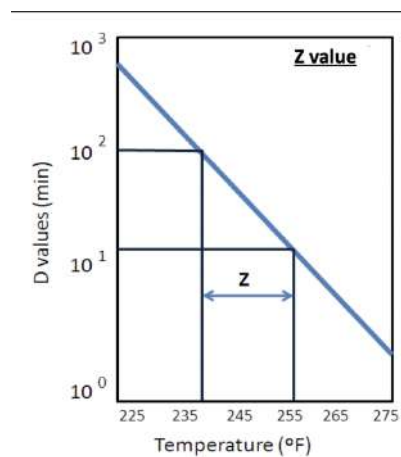


Figure 27: Z-Value Curve

13. Thermal Death Time (F-Value)

- Denoted as F_T

- **The Time must be used** to **Kill a Fixed Given Population of Organisms** at **Specified Temperature**
⇒ Targeted Reduction : Eliminate all of the Bacteria in that given population
- Function : Show How Heat Resistance Changes at differ Temperature
- Formula

$$\text{F-Value} = \text{D-Value} \times \text{Number of log}$$

$$\Rightarrow 1\text{-Log} = \text{D-Value}$$

- Unit : Time

14. Physical Methods

15. Heat Treatments I : Moist Heat (Pasteurization)

- (a) A Mild Heat Treatment
- (b) Function 1 : Inactivates Spoilage-Causing Enzymes
- (c) Function 2 : Kill Spoilage Non Spore Former Microorganisms
- (d) Type of Pasteurization
 - Low Temperature Long Time (LTLT) : **62°C + 30 Minutes**
 - High Temperature Short Time (HTST) : **72°C + (15 – 17) Seconds**
 - Ultra High Temperature (UHT) : **120°C – 138°C + (2 – 4) Seconds**
- (e) Pasteurization → Rapid Cooling
⇒ Prevent Microorganisms from Growing

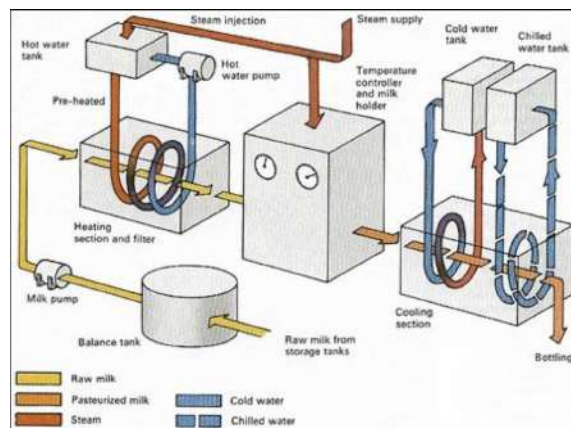


Figure 28: Pasteurization

16. Heat Treatments II : Moist Heat (Sterilization)

- (a) Commercial Sterilization : All Canned Foods sterilized in a **retort (Pressure Cooker)**
 $\Rightarrow$ Prove there is no viable organisms are present
- (b) Food with $\text{pH} > 4.6$: Heated under Pressure to 121°C for **Several Time**
 $\Rightarrow$ Destroy all Pathogenic & Spoilage Microorganisms
- (c) Food with **Spore Form** of *Clostridium botulinum* : Use Harmful Condition to destroy
 $\because$ It is Anaerobic + Can produce **Acute Botulinum Toxin**
 $\rightarrow$ Illness + Death
- (d) $\uparrow$ Pressure, $\uparrow$ Efficiency of Sterilization

17. Heat Treatments III : Microwave Sterilization

- (a) Food is exposed to **Radio Frequency (1-500 MHz) / Microwave (500 MHz - 10 GHz)**
- (b) Electric Dipole : Heat is generated by **Rapid Oscillation of Water Molecules**
 $\Rightarrow$ Keep Rotating Up-Side-Down in Electric Field
- (c) δ^- on Oxygen Part
- (d) δ^+ on Hydrogen Part

- (e) The molecule movement create **Intermolecular Friction** → Produce Heat directly to food
- (f) **Cannot use this method when the food contain less water**
∴ Cannot Rotate Water Molecules

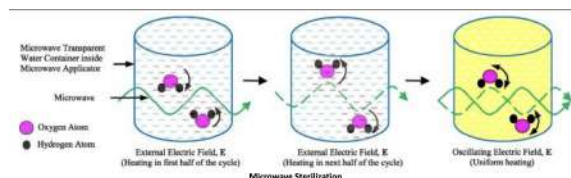
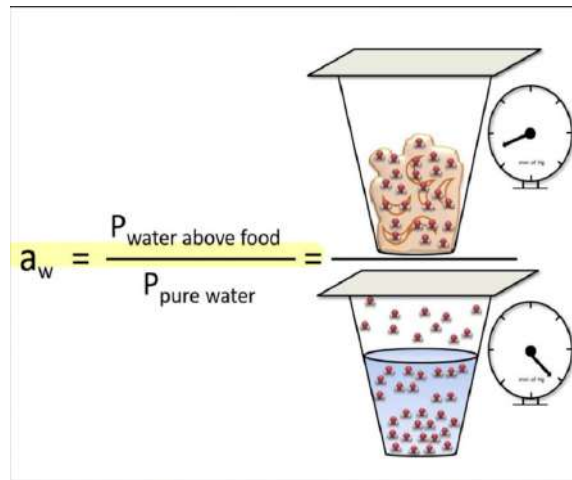


Figure 29: Microwave Sterilization

- (g) Heat is generate & transfer equally in the oven with **Faster heating Rate + Shorter Processing Time**
⇒ Traditional Heating : Heat is transfer from surface to the interior
- (h) Colour & Texture are better than traditional retorted foods
- (i) Food itself does not get heat uniformly → Uneven Lethality
- (j) **Cannot use this method when the food water is not evenly distributed**
⇒ Use microwave need to look at the matrix

18. Dehydration / Drying

- (a) Hot air Remove Water by Evaporation
⇒ Concentrated Form of Food is Remain
⇒ Nutrients is also retain
- (b) Drying Temperature & Reduced Water Activity : Delay / Prevent Bacterial Growth
- (c) Critical a_w = Minimum a_w

Figure 30: Dehydration with a_w

19. Freezing

- (a) Temperature : $< -18^{\circ}\text{C}$
 - $\Rightarrow$ Stops the Metabolic Activity
 - $\Rightarrow$ E.g. Inactivates Enzymes, Denature Proteins, Inhibit Metabolic Processes $\rightarrow$ No Deterioration
- (b) Produce Osmotic Shock in Microbes
 - $\Rightarrow$ Meaning **Cannot Use Water**
- (c) Intercellular Ice Crystal Formation $\rightarrow$ Mechanical Injury (Pressure)
 - $\Rightarrow$ Cell Membrane suffer major damage
- (d) Storage Life Time : Months - Years

20. Thawing

- Cannot be Long
- Damage Food Cell $\rightarrow$ Release Nutrient Rich Liquid
- Cold Survival Pathogen can multiply

21. Freezing : Freeze Drying

- (a) Freeze-Drying = Lyophilization
 - $\Rightarrow$ Combining Freezing & Drying (Dehydration)
- (b) Dehydration : Cause by Vacuum Sublimation of the ice

- (c) Sublimation : Water move from **Solid State → Gas State**
⇒ Gentle Way to Remove Water
- (d) Solute Nutrient Remain Inside the Food
- (e) Food Structure is also retains

22. Changing The Atmosphere I : Cool Storage

- (a) Cool Storage = Chilled Storage
- (b) Temperature Range : -2 to 16°C
- (c) Time : Few Days - Several Weeks
- (d) $\downarrow$ Temperature $\rightarrow \downarrow$ Metabolic Reactions

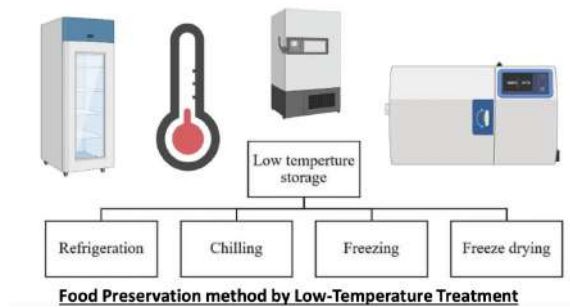
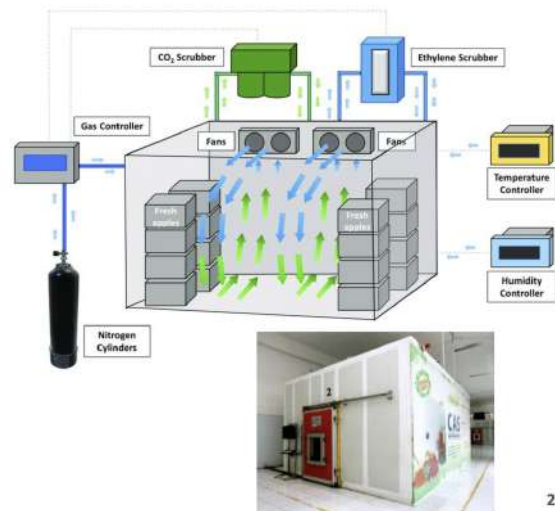


Figure 31: Low Temperature Treatment

23. Changing The Atmosphere II : Control Atmosphere Storage (CAS)

- (a) Used for Fruits & Vegetable
- (b) O_2 Level : **$21\% \rightarrow 0.5\% - 2.5\%$**
- (c) CO_2 Level : **$0.045\% \rightarrow 8\% - 10\%$**
- (d) $\uparrow \text{CO}_2 \rightarrow \downarrow$ Microbial Growth
⇒ Need $\downarrow$ Respiration + $\downarrow$ pH



27

Figure 32: Control Atmosphere Storage (CAS)

24. Changing The Atmosphere III : Modified Atmosphere Packaging (MAP)

(a) Control the Composition of Gases

(b) Gaseous Mixture : O_2, CO_2, N_2

25. Radiation I : UV Radiation

(a) UV Radiation = Non-Ionizing = 240 – 280 nm

⇒ This wavelength is destructive

(b) Damage Nucleic Acids

⇒ Use Cross-Linking thymine Dimer in DNA

(c) Prevent Repair + Reproduction

(d) Can kill Gram-Negative Bacteria

⇒ Cannot kill Bacterial Endospore & Molds & Virus

∴ They are more Resistant

(e) Penetrating Power is Weak

⇒ Highly Depending on the Concentration of Solute in Food

⇒ Cannot kill all bacteria

(f) Mainly Used in Surface Disinfection

26. Radiation II : Ionizing Radiation

- (a) X-Ray, Gamma-Ray
- (b) Strong Penetration Power
- (c) Can Kill Microbes by Damaging DNA
⇒ Not Affecting the quality of Food
- (d) Sensitivity : Gram-Negative Bacteria, Spoilage Organisms, Pathogenic Species > Gram-Positive Bacteria

27. Novel Physical Method for Preservation

28. Dense-Phase Carbon Dioxide (DPCD)

- (a) Non-Thermal Processing
- (b) Usually Use at **Liquid Sample**
- (c) Use High Pressure CO₂
⇒ Pressure Level : Supercritical State
⇒ Above Critical Point
- (d) High Pressure → Change CO₂ into Liquid Phase → Mix Liquid CO₂ with Food Together → Release Pressure below Critical Point → Change CO₂ back to **Gas State** → MO Burst
- (e) Liquid CO₂ : ↑ Thermodynamic Stability + Eliminate Flammability & Toxicity
- (f) Liquid Food : Less Effective to Solid Food
⇒ Treatment : Mild Temperature (30 – 50°C) + Pressure above Critical Point of CO₂ → CO₂ Injected

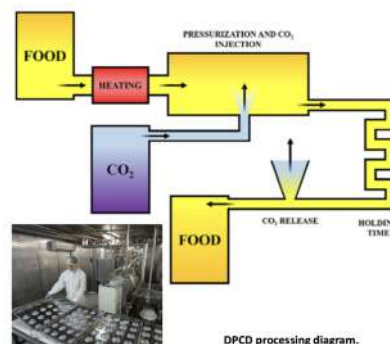


Figure 33: DPCD Processing Diagram

- (g) CO₂ Addition : ↓ pH of Liquid Food Medium + **Denaturation** of Microbial Enzymes + **Form Bicarbonate Complex with Enzymes** + Precipitation of Metal Ions with Enzymes
 ⇒ Metal Ions + Enzyme → Form Biocarbonate Salts
- (h) Release of CO₂ : Depressurization
 ⇒ Expansion of CO₂ → Cell Lysis + Release Vital Biomolecules
 (Also Harmful to Cell Viability)

29. Cold Plasma

- (a) Plasma : A Gas which have Ionized Molecules & Atom
 ⇒ Forth State of Matter
- (b) Plasma can come from **Different Gases**
 ⇒ E.g. Nitrogen, Oxygen, Mixture of Oxygen & Nitrogen, Noble Gases
- (c) Cold Plasma : Plasma that can generated at **Room Temperature**
- (d) **Collision of Plasma Inactivates Microorganisms & Essential Biomolecules** → UV Radiation Release (High Energy)
 ⇒ Essential Biomolecules : Genetic Material, Protein
 ⇒ Cell Membranes are Degraded

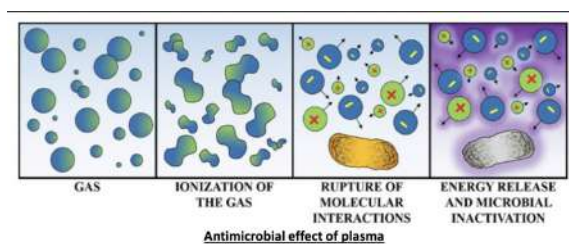


Figure 34: Cold Plasma

30. Food Antimicrobial

- Chemical Added to Food
- Used to Inhibit Growth, Kill Microorganisms
- Normal Used Concentration : **Bacteriostatic & Fungistatic**
 ⇒ Inhibit Bacteria & Fungi Growth
 ∴ Food Antimicrobial Cannot Preserve Food Indefinitely

- Not Bactericidal & Fungicidal
⇒ Kill Bacteria & Fungi
- Common Name : Preservatives
- Antimicrobial Activity Affecting Factor : Microbiological, Intrinsic, Extrinsic, Process Factors

31. Mechanisms of Natural Antimicrobial Preservatives

- Inhibition of Cell Wall Synthesis
- Disruption of Plasma Membrane
- Interference with Metabolite Synthesis
- Inhibition of Protein Synthesis
⇒ Inhibit Translation Process
- Inhibition of RNA Synthesis
⇒ Inhibit Transcription Process
- Inhibition of DNA Replication

32. Any of the above is Antimicrobial ⇒ Can Inhibit Bacteria & Fungi

Microbiological	Intrinsic	Extrinsic	Process
Genetic resistance	pH	Duration of storage	Alteration of food microstructure
Vegetative cell or spore	Redox potential	Temperature	Alteration of microbial ecology
Cell number	a_w	Gaseous atmosphere	Alteration of food composition
Growth phase	Buffering capacity	Relative humidity	Induced responses, such as injury
Growth rate	Food composition	Packaging environment	
Inactivation mechanisms			
Interaction with other microbes			
Cell physiology			

Figure 35: Antimicrobial Activity Factors

33. Natural Occurring Antimicrobial

- Extracted from Natural Source
- No Side Effects
- Good for Consumers
- Organic

- Examples : Acetic Acid (Vinegar), Benzoic Acid (Cranberries), Sorbic Acid (Rowan Berries)

34. Natural Occurring Antimicrobial I : **Lactoferrin & iron Binding Proteins**

- (a) Iron Stimulates Growth
⇒ E.g. *Clostridium*, *Escherichia*, *Listeria*, *Pseudomonas*, *Salmonella*, *Staphylococcus*
- (b) Iron Binding Protein : Milk, Eggs
⇒ E.g. **Lactoferrin (Common in Milk)**, **Transferrin (Low Level in Milk)**, **Ovotransferrin (Egg Albumin = Egg White)**
- (c) Limitation : ↓ Iron Usage of Microbes
- (d) Gram-Positive Bacteria : More Sensitive to Iron
⇒ E.g. *Bacillus*, *Micrococcus*

35. Natural Occurring Antimicrobial II : **Avidin**

- (a) Egg Albumin (Egg White) Protein
⇒ 0.05% of Total Albumin Protein
- (b) Stable to Heat
- (c) Wide Range of pH
- (d) Pathway 1 : **Avidin Binds Biotin** → Inhibit Biotin Required Bacteria & Yeast
- (e) Pathway 2 : **Avidin Binds Transport Protein in *E.coli* Outer Membrane** → Interfering *E. coli* Transportation

36. Natural Occurring Antimicrobial III : **Spices & Essential Oils**

- (a) Spices : Flavoring agents
- (b) Interfering **Membrane Function** of Microbes

37. Natural Occurring Antimicrobial IV : **Onion & Garlic**

- (a) Plant Antimicrobial
- (b) Inhibit Bacteria & Fungi Growth
- (c) Garlic Antimicrobial : Allicin

- Bind with S Compound (Sulfur)
- Inhibit Sulfhydryl Containing Enzymes in Bacteria
- (d) Onion Antimicrobial I : Allicin
 - Inhibit Sulfhydryl Containing Enzymes in Bacteria
 - Forming Thiopropanal-S-oxide
- (e) Onion Antimicrobial II : Phenolic Compound
 - Influence Cell Membrane Function

38. Natural Occurring Antimicrobial V : Isothiocyanates

- (a) Chemical Formula : $\mathbf{R-N=C=S}$
- (b) Strong Potent Antimicrobial
- (c) Made from **Glucosinolates (In Plant Cells : From Cruciferae)**
 $\Rightarrow$ E.g. Cabbage, Mustard, Brussels, Broccoli, Kale
- (d) Can Inhibit Cells by **Disulfide Bonds & Inactivating Sulfhydryl Enzymes**

39. Chemical Antimicrobial

- Synthesized from Natural Compounds
- Might have Side Effects
- Bad for Consumers

40. Antimicrobial Chemicals I : Organic Acids & Derivatives

- (a) Uncharged Form = Protonated Form $\rightarrow$ Can Migrate into Cell Membrane Easily
- (b) $\text{pH} < \text{pKa}$: Protonated Form is Dominated
- (c) Only Use this Method when $\text{pH} < \text{pKa}$
 $\Rightarrow \text{pH} > \text{pKa}$: Ionize Form Compounds cannot enter Bacteria Cells $\rightarrow$ Cannot Use
- (d) Short Chain Organic Acids
- (e) $\text{pH} : \leq 5.5$
- (f) pKa : Maximum pH of the certain pH level that the antimicrobial is on / off

(g) Mechanism : Protonated Form Molecules $\xrightarrow[\text{Internal pH} = 7.0 > \text{pKa}]{\text{Cross Cell membrane}}$

Release Proton $\xrightarrow{\text{ATPase}}$ Pump Extra H^+ out the Cell Membrane

$\Rightarrow$ ATPase : Use Energy

$\Rightarrow$ Keep Pumping Out $\rightarrow$ Use All ATP

$\Rightarrow$ Bacteria Died as **ATP is Shortage** $\rightarrow$ Cannot Support Bacterial Growth & Maintenance

(h) Chemical Formula : $\text{HA} (50\%) \rightleftharpoons \text{H}^+ + \text{A}^- (50\%)$

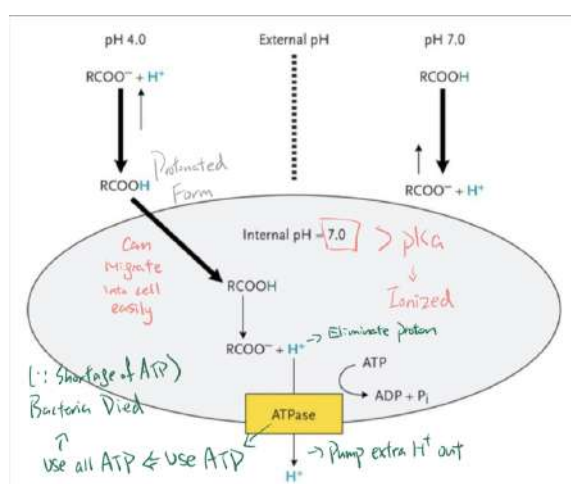


Figure 36: H^+ Pathway

41. Antimicrobial Chemicals II : Parabenoic Acids

(a) Common Name : Parabens

(b) Effective pH Range : 3.0 – 8.0

$\Rightarrow$ Assume it is Remain Intact Form & Undissociated Form

(c) Only Useful when $\text{pH} \leq \text{pKa} = 8.5$

(d) Mainly Used at Baked Food & Beverages & Fruit Products & Jams

$\Rightarrow$ E.g. Methyl, Propyl, Heptyl

(e) **How Strong Parabens is Directly Proportional to Alkyl Chain Length**

$\Rightarrow \uparrow$ Alkyl Length, $\uparrow$ Inhibitory Activity

(f) Can **Damage Membrane Barrier**
 $\Rightarrow$ Intracellular Compounds Leak Out

(g) Can **Inhibit Respiration**
 $\therefore$ Changing the Membrane Fluidity

42. Antimicrobial Chemicals III : Nitrites (NaNO_2 & KNO_2)

(a) Most Common Antimicrobial Chemical

(b) Usually Used in Cured Meats
 $\Rightarrow$ E.g. Hot Dogs, Bacon

(c) React with **Muscle Protein : Myoglobin** $\rightarrow$ Form **Nitrosomyoglobin**
 $\Rightarrow$ Make Pink Colour to the Cured Meat

(d) Used at Low pH Food (Acidic) + Lack of Air

(e) Anaerobic Environment : Stronger Inhibition than Aerobic Environment

(f) Nitrosamines : Formed by Nitrite

(g) Amines are Carcinogens
 $\therefore$ Have Health Issue Concern

43. Antimicrobial Chemicals IV : Sodium Chloride (NaCl)

(a) Salting : Dehydrate Food

(b) Can Combine with other Processes
 $\Rightarrow$ E.g. Canning, Pasteurization, Drying, Dehydration

(c) Lethal Mechanism : **Plasmolysis**
 $\Rightarrow$ Water Keep Moving Out (Water Loss)

(d) Salt Reduce Water Activity $\rightarrow$ Osmotic Shock $\rightarrow$ Create Unfavorable Growth Condition to Cells

44. Antimicrobial Chemicals V : Phosphates

(a) Polyphosphates : Use **Chelate Essential Cation from Cell Wall Binding Sites** to Kill Microorganisms
 $\Rightarrow$ E.g. K^+ Channel Sites, Na^+ Channel Sites

(b) Affected by Phosphates : Gram-Positive Bacteria $>$ Gram-Negative Bacteria

45. Antimicrobial Chemicals VI : Sulfites

- (a) A Disinfectant
- (b) Used in Fruits, Vegetable → Control Spoilage Yeast & Molds
+ Control Acetic Acid bacteria + Control Malolactic Bacteria
- (c) Act on Microbes Cytoplasmic Membrane, DNA replication, Protein Synthesis (Translation)
- (d) Have Antioxidant Properties
- (e) Function : Inhibit Browning

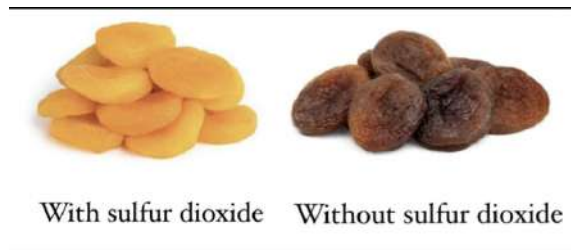


Figure 37: Browning